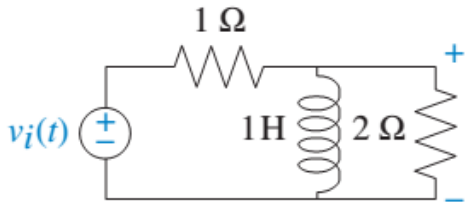


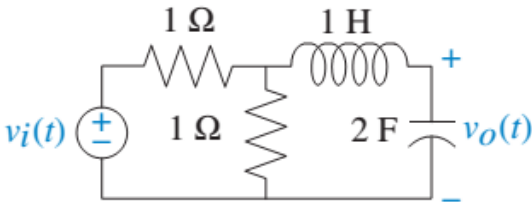
Assignment 4 (ELEC 341 L4_Model_ElecMech)

Problem 1:

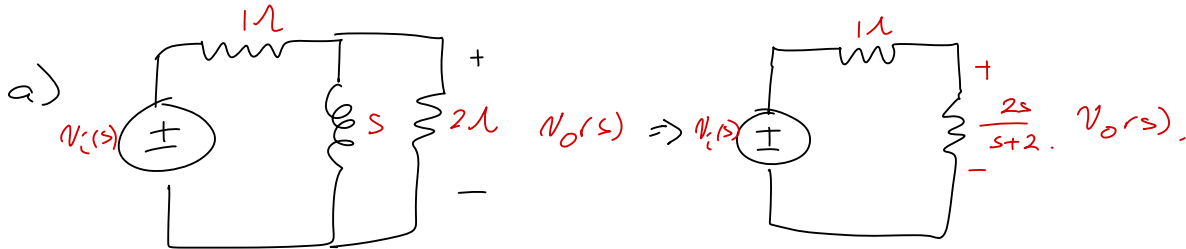
Find the transfer function, $G(s) = V_o(s)/V_i(s)$ for each network shown in the following figure:



(a)



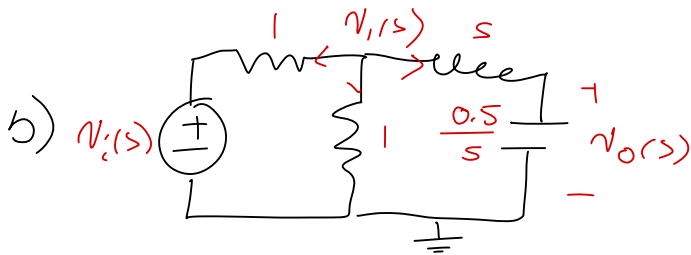
(b)



$$V_o(s) = V_i(s) \cdot \frac{\frac{2s}{s+2}}{1 + \frac{2s}{s+2}} \Rightarrow$$

$$G(s) = \frac{\frac{2s}{s+2}}{1 + \frac{2s}{s+2}} = \frac{2s}{s+2+2s}$$

$$G(s) = \frac{2s}{3s+2}$$



$$\textcircled{1} \quad V_1(s) - V_i(s) + V_1(s) + \frac{V_1(s) - V_o(s)}{s} = 0 \Rightarrow 2sV_1(s) - sV_i(s) + V_1(s) - V_o(s) = 0$$

$$(2s+1)V_1(s) - sV_i(s) - V_o(s) = 0$$

$$\textcircled{2} \quad \frac{V_1(s) - V_o(s)}{s} = \frac{V_o(s)}{\frac{0.5}{s}} \Rightarrow \frac{V_1(s) - V_o(s)}{s} = 2sV_o(s)$$

$$= V_1(s) - V_o(s) = 2s^2V_o(s)$$

$$V_1(s) = (2s^2+1)V_o(s)$$

$$(2s+1)(2s^2+1)V_o(s) - sV_i(s) - V_o(s) = 0$$

$$V_o(s) \left((2s^2+1)(2s+1) - 1 \right) = sV_i(s)$$

$$G(s) = \frac{s}{(2s^2+1)(2s+1) - 1} = \frac{1}{4s^2+2s+2}$$

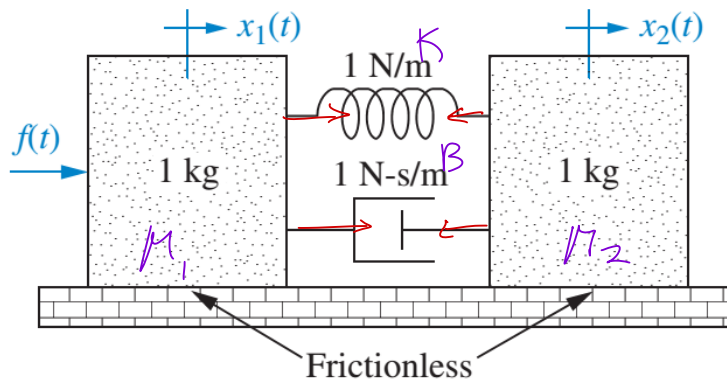
$$(2s^2+1)(2s+1)-1 = 4s^3+2s^2+2s+1$$

Assignment 4 (ELEC 341 L4_Model_ElecMech) $= s(4s^2+2s+2)$

Problem 2:

draw forces away from mass.

Find the transfer function, $G(s) = X_2(s)/F(s)$, for the translational mechanical network shown in figure:



mass 1 $M_1 x_1'' = -K(x_1 - x_2) - B(x_1' - x_2') + f(t)$

$$x_1'' + x_1' + x_1 = x_2' + x_2 + f(t)$$

$$s^2 X_1(s) + s X_1(s) + X_1(s) = s X_2(s) + X_2(s) + F(s)$$

$$\textcircled{1} X_1(s)(s^2 + s + 1) = s X_2(s) + X_2(s) + F(s)$$

mass 2 $M_2 x_2'' = K(x_1 - x_2) + B(x_1' - x_2')$

$$x_2'' + x_2' + x_2 = x_1' + x_1$$

$$s^2 X_2(s) + s X_2(s) + X_2(s) = s X_1(s) + X_1(s)$$

$$X_2(s)(s^2 + s + 1) = (s + 1) X_1(s)$$

$$\textcircled{2} X_1(s) = \frac{(s^2 + s + 1)}{s + 1} X_2(s)$$

combine eqn 1 and 2

$$\frac{(s^2 + s + 1)(s^2 + s + 1)}{s + 1} X_2(s) = X_2(s)(s + 1) + F(s)$$

$$\left[\frac{(s^2 + s + 1)^2}{s + 1} - (s + 1) \right] X_2(s) = F(s)$$

$$G(s) = \frac{X_2(s)}{F(s)} = \frac{s + 1}{(s^2 + s + 1)^2 - (s + 1)^2}$$

$$\boxed{G(s) = \frac{s + 1}{s^2(s^2 + 2s + 2)}}$$

Assignment 4 (ELEC 341 L4_Model_ElecMech)

Problem 3:

Calculate the transfer function of the system represented in the state space model as

$$\dot{X} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} [u]$$
$$Y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \overset{\sim}{0}$$

$$\tilde{A} = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\tilde{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\tilde{C} = (0 \ 1)$$

$$\tilde{D} = \overset{\sim}{0}$$

$$H(s) = C(sI - A)^{-1}B + D$$

$$= (0 \ 1) \begin{pmatrix} s+1 & 1 \\ -1 & s \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\frac{1}{s(s+1)+1} \begin{pmatrix} s & -1 \\ 1 & s+1 \end{pmatrix}$$

$$= \frac{(0 \ 1)_{1 \times 2}}{s^2+s+1} \begin{pmatrix} s & -1 \\ 1 & s+1 \end{pmatrix}_{2 \times 2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}_{2 \times 1}$$

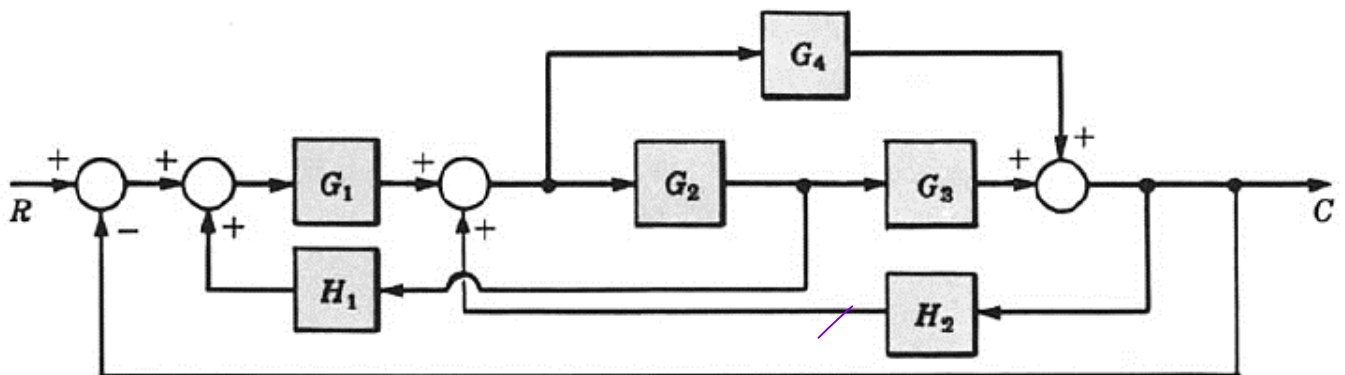
$$= \frac{1}{s^2+s+1} (1 \ s+1)_{1 \times 2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}_{2 \times 1}$$

$$H(s) = \frac{1}{s^2+s+1}$$

Assignment 4 (ELEC 341 L4_Model_ElecMech)

Problem 4:

Find the signal flow graph for the following block diagram:



12:53 PM Tue May 26

